

MULTIPLE SEQUENCE ALIGNMENT REPORT

Title:

Multiple Sequence Alignment of TP53 Protein Across Mammalian Species

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1. Introduction

The TP53 gene encodes the tumor suppressor protein p53, a critical regulator of the cell cycle, DNA repair, apoptosis, and genomic stability. Due to its essential role in preventing cancer development, TP53 is one of the most studied genes in molecular biology.

Multiple Sequence Alignment (MSA) is a computational method used to compare biological sequences (DNA or proteins) across different species to identify conserved regions, functional domains, and evolutionary relationships. Highly conserved residues often indicate structurally or functionally important regions that are maintained through evolutionary pressure.

This study performs a multiple sequence alignment of TP53 protein sequences across selected mammalian species to evaluate conservation patterns and infer evolutionary relationships.

2. Materials and Methods

2.1 Sequence Retrieval

Protein sequences of TP53 were retrieved from the UniProt database in FASTA format. Sequences were selected from five mammalian species:

- Homo sapiens (Human)
- Pan troglodytes (Chimpanzee)
- Gorilla gorilla (Gorilla)
- Mus musculus (Mouse)
- Rattus norvegicus (Rat)

Both reviewed (Swiss-Prot) and unreviewed (TrEMBL) entries were included where appropriate due to limited availability of fully reviewed orthologs for some species.

2.2 Multiple Sequence Alignment Tool

Multiple Sequence Alignment was performed using **Clustal Omega**, a widely used web-based alignment tool provided by the European Bioinformatics Institute (EBI).

Website: <https://www.ebi.ac.uk/Tools/msa/clustalo/>

2.3 Procedure

1. TP53 protein sequences were collected from UniProt in FASTA format.
2. All sequences were compiled into a single FASTA file.

- 3. The FASTA file was uploaded into Clustal Omega.
- 4. Multiple sequence alignment was executed using default parameters.
- 5. The resulting alignment was analyzed for conserved residues, gaps, and substitution patterns.
- 6. Output visualization and alignment screenshots were saved for interpretation.

3. Results

The multiple sequence alignment revealed strong conservation across all five mammalian TP53 sequences.

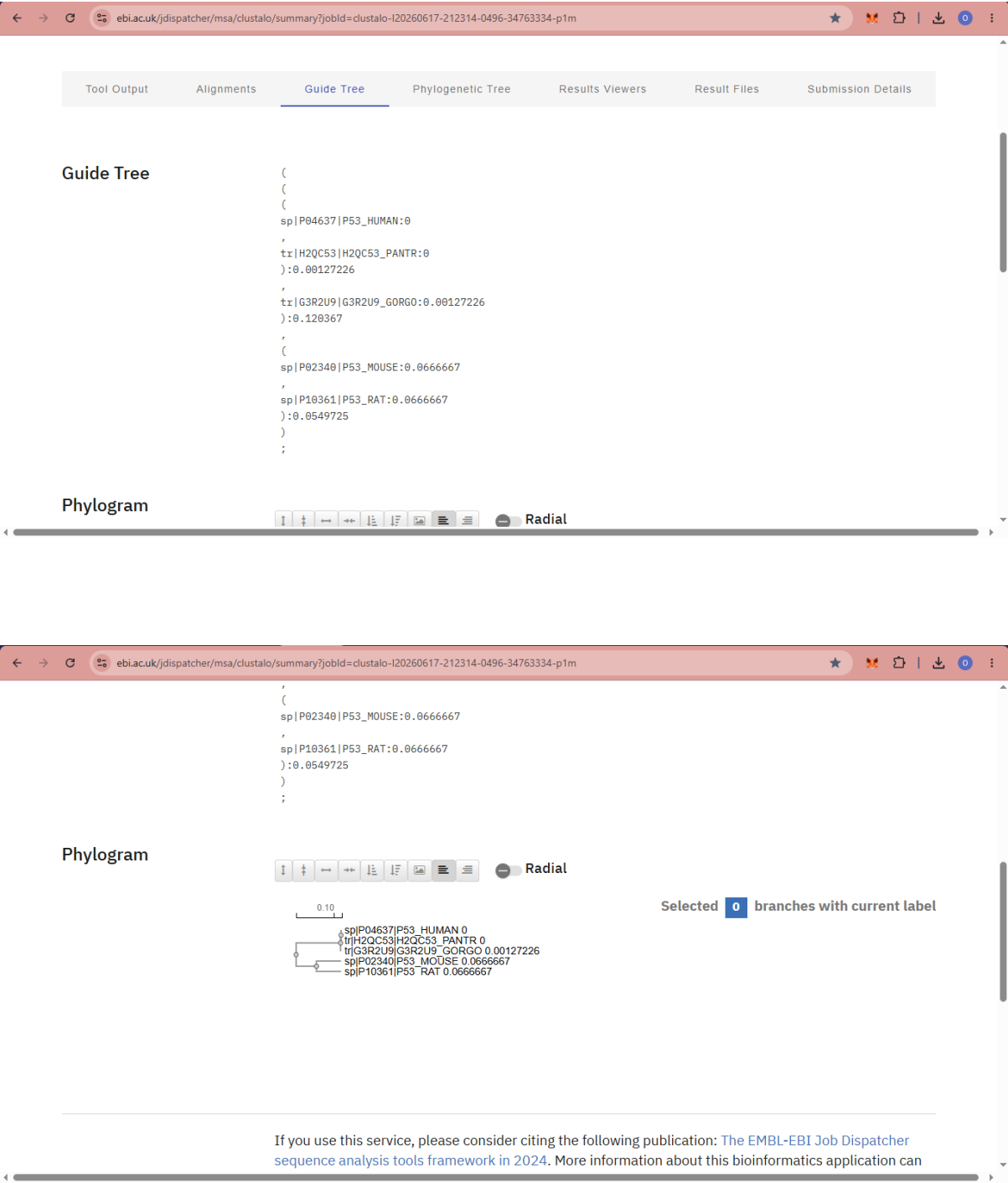
Key observations include:

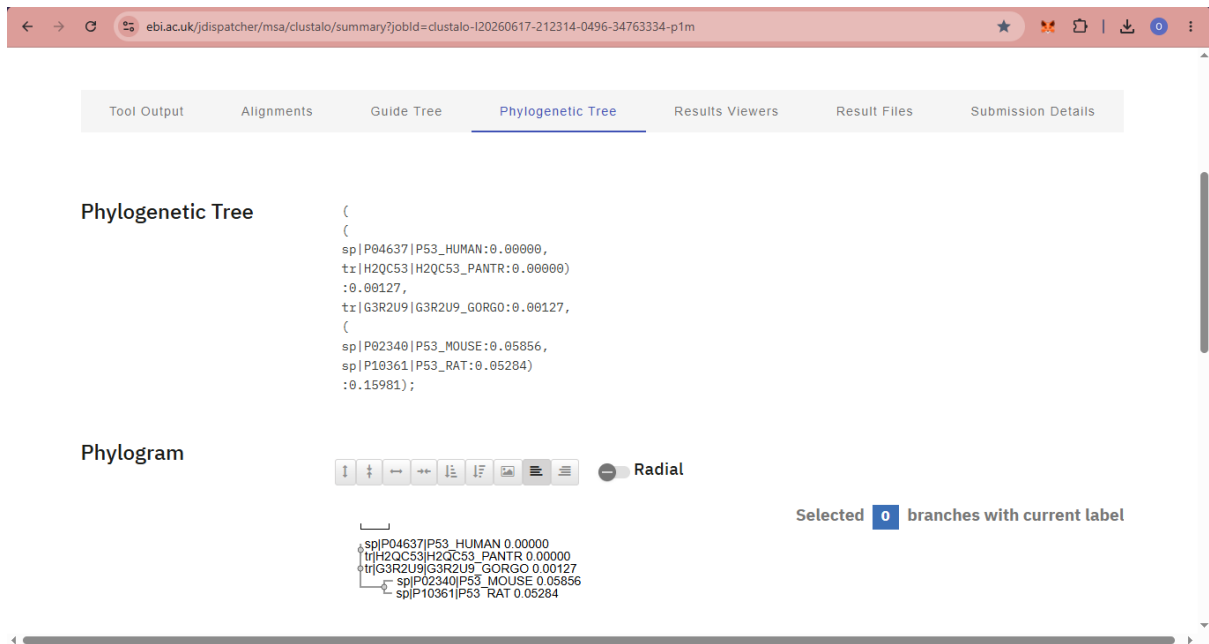
- High sequence identity among primates (human, chimpanzee, and gorilla), with near-identical amino acid sequences across most regions.
- Moderate divergence observed in rodent sequences (mouse and rat), particularly in non-core regions.
- The central DNA-binding domain of TP53 exhibited the highest level of conservation across all species.
- Conserved residues were indicated by asterisks (*), showing identical amino acids across all sequences.

Figure 1: Multiple Sequence Alignment of TP53 Protein (Clustal Omega Output)



Figure 3: Multiple Sequence Alignment of TP53 Protein (Guide Tree and Phylogenetic Tree)





4. Interpretation

The multiple sequence alignment of TP53 proteins across mammalian species demonstrates strong evolutionary conservation, particularly within the central DNA-binding domain, which is essential for the tumor suppressor function of p53.

Primates (human, chimpanzee, and gorilla) show extremely high sequence similarity, with only minor amino acid substitutions. This reflects their recent evolutionary divergence and the strong functional constraints acting on TP53.

In contrast, rodent species (mouse and rat) exhibit a higher degree of variation, including amino acid substitutions and gaps. Despite this divergence, key functional residues remain conserved, indicating strong selective pressure to maintain the biological activity of the protein.

Overall, the results suggest that TP53 is highly conserved across mammals due to its essential role in regulating cell cycle control and preventing tumor formation. The observed conservation pattern is consistent with expected evolutionary relationships among the selected species.

5. Conclusion

The TP53 protein is highly conserved across mammalian species, particularly within its functional domains. The multiple sequence alignment highlights evolutionary conservation among primates and moderate divergence in rodents, reflecting phylogenetic relationships.

These findings reinforce the critical biological importance of TP53 in maintaining genomic integrity.

6. References

- UniProt Consortium. UniProt: a worldwide hub of protein knowledge.
<https://www.uniprot.org/>
- Sievers et al. (2011). Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega.
- European Bioinformatics Institute (EBI): <https://www.ebi.ac.uk/Tools/msa/clustalo/>